

# MQTT commads functions report

## Introduction

The MQTT service allows UWP 4.0 to receive commands in order to control the configured functions. To enable the MQTT service to receive commands for a specific function, you need to select the relevant function in the web app. A MQTT client has to publish a command in the set "command topic" to control a function.

The payload is in JSON format and has this structure:

- **id:** the function ID that needs to receive the command (mandatory, integer)
- **cmd:** the command to be sent to the function, written in literal version (mandatory, string, see list)
- **value:** the value that can be assigned to the command; it can be a timer, a temperature, a percentage (optional, int or double)
- **values:** allows you to send a command to set more parameters. Only specific function supports this feature. For example, a fx Motor where you can define position and tilt angle (optional, string).
- **topic:** this property has to be defined only if the Response Topic has been set on the broker side. This is the target topic that the service will use to send the response of successful command or related errors (optional , string).

### Example:

```
{"id":1, "cmd": "SetValue", "value":5, "topic": "responseTopic" }
```

### Response:

When the Response Topic is set on the broker side, these are examples of response messages that can be sent to MQTT clients:

In case of valid command, a successful response is sent:

```
{"code": 0, "msg": "ok" }
```

In case of command sending or execution errors, an error response is sent:

```
{"code": 7, "msg": "Invalid function" }
```

### Error codes table:

| Error code: | Error message:       | Description:                            |
|-------------|----------------------|-----------------------------------------|
| 7           | Missing function id  | Missing function ID for payload         |
| 8           | Missing command      | Missing command for payload             |
| 9           | Invalid json command | Wrong json format for payload           |
| 10          | Invalid function     | Function is invalid                     |
| 11          | Invalid function id  | Function ID is not in the selected list |
| 12          | Invalid command      | Command is not valid                    |

**Functions list and their commands:**

| Counter       | id   |
|---------------|------|
| Smoke Status  | 3549 |
| Temperature   | 3585 |
| PLC - SIEMENS | 6563 |

| Command              | Payload                                          |
|----------------------|--------------------------------------------------|
| Increase             | {"id": <id>, "cmd": "Increase", "value":<value>} |
| Decrease             | {"id": <id>, "cmd": "Decrease", "value":<value>} |
| SetValue             | {"id": <id>, "cmd": "SetValue", "value":<value>} |
| ResetValue           | {"id": <id>, "cmd": "ResetValue"}                |
| Freeze               | {"id": <id>, "cmd": "Freeze"}                    |
| Unfreeze             | {"id": <id>, "cmd": "Unfreeze"}                  |
| FreezeUnfreezeToggle | {"id": <id>, "cmd": "FreezeUnfreezeToggle"}      |
| ResetRollover        | {"id": <id>, "cmd": "ResetRollover"}             |

| Alarm | id   |
|-------|------|
| Alarm | 3667 |

| Command            | Payload                                   |
|--------------------|-------------------------------------------|
| Acknowledgement    | {"id": <id>, "cmd": "Acknowledgement"}    |
| Reset              | {"id": <id>, "cmd": "Reset"}              |
| ResetOn            | {"id": <id>, "cmd": "ResetOn"}            |
| ResetOnTimeout     | {"id": <id>, "cmd": "ResetOnTimeout"}     |
| ResetOff           | {"id": <id>, "cmd": "ResetOff"}           |
| ResetToggle        | {"id": <id>, "cmd": "ResetToggle"}        |
| ResetToggleTimeout | {"id": <id>, "cmd": "ResetToggleTimeout"} |
| TestAlarmOn        | {"id": <id>, "cmd": "TestAlarmOn"}        |
| RemoveTestAlarmOn  | {"id": <id>, "cmd": "RemoveTestAlarmOn"}  |
| TestAlarmOnToggle  | {"id": <id>, "cmd": "TestAlarmOnToggle"}  |
| TestAlarmOff       | {"id": <id>, "cmd": "TestAlarmOff"}       |
| RemoveTestAlarmOff | {"id": <id>, "cmd": "RemoveTestAlarmOff"} |
| ResetToggleTimeout | {"id": <id>, "cmd": "ResetToggleTimeout"} |

| Zone temperature | id |
|------------------|----|
|------------------|----|

| Zone temperature | id   |
|------------------|------|
| Zone temperature | 4147 |

| Command                   | Payload                                                          |
|---------------------------|------------------------------------------------------------------|
| HeatingActivation         | {"id": <id>, "cmd": "HeatingActivation"}                         |
| HeatingDeactivation       | {"id": <id>, "cmd": "HeatingDeactivation"}                       |
| HeatingToggleActivation   | {"id": <id>, "cmd": "HeatingToggleActivation"}                   |
| HeatingSetPointSelection  | {"id": <id>, "cmd": "HeatingSetPointSelection", "value":<value>} |
| HeatingSetS1              | {"id": <id>, "cmd": "HeatingSetS1", "value":<value>}             |
| HeatingSetS2              | {"id": <id>, "cmd": "HeatingSetS2", "value":<value>}             |
| HeatingSetS3              | {"id": <id>, "cmd": "HeatingSetS3", "value":<value>}             |
| HeatingOffset             | {"id": <id>, "cmd": "HeatingOffset", "value":<value>}            |
| HeatingFanSpeedMode       | {"id": <id>, "cmd": "HeatingFanSpeedMode", "value":<value>}      |
| HeatingActivateForceOn    | {"id": <id>, "cmd": "HeatingActivateForceOn"}                    |
| HeatingDeactivateForceOn  | {"id": <id>, "cmd": "HeatingDeactivateForceOn"}                  |
| HeatingToggleForceOn      | {"id": <id>, "cmd": "HeatingToggleForceOn"}                      |
| HeatingActivateForceOff   | {"id": <id>, "cmd": "HeatingActivateForceOff"}                   |
| HeatingDeactivateForceOff | {"id": <id>, "cmd": "HeatingDeactivateForceOff"}                 |
| HeatingToggleForceOff     | {"id": <id>, "cmd": "HeatingToggleForceOff"}                     |
| CoolingActivation         | {"id": <id>, "cmd": "CoolingActivation"}                         |
| CoolingDeactivation       | {"id": <id>, "cmd": "CoolingDeactivation"}                       |
| CoolingToggleActivation   | {"id": <id>, "cmd": "CoolingToggleActivation"}                   |
| CoolingSetPointSelection  | {"id": <id>, "cmd": "CoolingSetPointSelection", "value":<value>} |
| CoolingSetS1              | {"id": <id>, "cmd": "CoolingSetS1", "value":<value>}             |
| CoolingSetS2              | {"id": <id>, "cmd": "CoolingSetS2", "value":<value>}             |
| CoolingSetS3              | {"id": <id>, "cmd": "CoolingSetS3", "value":<value>}             |
| CoolingOffset             | {"id": <id>, "cmd": "CoolingOffset", "value":<value>}            |
| CoolingFanSpeedMode       | {"id": <id>, "cmd": "CoolingFanSpeedMode", "value":<value>}      |
| CoolingActivateForceOn    | {"id": <id>, "cmd": "CoolingActivateForceOn"}                    |
| CoolingDeactivateForceOn  | {"id": <id>, "cmd": "CoolingDeactivateForceOn"}                  |
| CoolingToggleForceOn      | {"id": <id>, "cmd": "CoolingToggleForceOn"}                      |
| CoolingActivateForceOff   | {"id": <id>, "cmd": "CoolingActivateForceOff"}                   |
| CoolingDeactivateForceOff | {"id": <id>, "cmd": "CoolingDeactivateForceOff"}                 |
| CoolingToggleForceOff     | {"id": <id>, "cmd": "CoolingToggleForceOff"}                     |
| DisableOn                 | {"id": <id>, "cmd": "DisableOn"}                                 |

|                      |                                             |
|----------------------|---------------------------------------------|
| DisableOnTimeout     | {"id": <id>, "cmd": "DisableOnTimeout"}     |
| DisableOff           | {"id": <id>, "cmd": "DisableOff"}           |
| DisableToggle        | {"id": <id>, "cmd": "DisableToggle"}        |
| DisableToggleTimeout | {"id": <id>, "cmd": "DisableToggleTimeout"} |